

Joint inversion of gravity and gravity gradient and its application to mineral exploration

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ABSTRACT

Airborne gravity gradiometry has brought a remarkable improvement in mineral and petroleum exploration due to its high resolution. Structures of very different sizes and densities can be imaged due to its high spatial frequency information with quick and consistent coverage over a wide range of terrains. However, even the accurate mappings of underground structures of complex mineral deposits in such terrains are still lacking in terms of knowing the precise size, shape and density properties to avoid the drilling cost. Gravity gradiometry data significantly enhances the high frequency information than classical/conventional gravity data and can be jointly used to provide more accurate and refined underground structures. In this study, we have developed an open-source MATLAB based algorithm to analyse the performance of joint inversion of gravity and gravity gradient data to map such mineral deposits using different polygons associated with the density value. The inverted model is constrained with three different approaches of compactness, location and density. However, the number of vertex points directly describes the surface refinement strategy adopted in the study. We performed our analysis using Particle Swarm Optimization (PSO) method and polygonal approximation for the synthetic as well as real data sets from three different natural resource deposits of salt dome, sulfide and rare Earth Elements (REE). The developed PSO based inversion algorithm does not consider any regularization and constraining parameter to converge the solution. The results clearly show an improvement due to joint inversion using PSO in the precision of density and geometrical parameters (location, shape, size, depth and density) of complex ore deposits from the real field examples.

Keywords: Joint inversion, Gravity and gravity gradiometry, Particle swarm optimization, Polygonal approximation, Mineral exploration

INTRODUCTION

With the increasing trend of societal needs and scientific developments, the scenario of near surface exploration for groundwater, minerals and energy resources is changing rapidly. The continuous exploration of natural resources from several decades forced us to look for these resources in the deeper subsurface, ocean resources and around mud volcanoes followed by coastal deposits for heavy minerals. The knowledge on the resources in the deeper subsurface is very little at present. In particular, explorations of natural resources due to the associated geometrical factors like shape, size, depth and density (Dubey, 2015). These parameters directly influence with the formation of ore deposits that exhibit high density contrasts between the host rock and overburden as shown in Figure 1. Figure 1 illustrates the configuration of different types of ore deposits viz. (a) platinum reefs or coal deposits in form of bed or flat terrain, (b) iron ore or sulphide deposits in the form of massive deposits, (c) anthracite deposits in dipping bed, (d) copper sulphide deposits in the form of massive deposits and high relief, and (e) iron ore or coal deposits in form of thick bedded with little overburden or flat terrain.

The ore deposits can be modelled at different depths using accurate physical (density) and geometrical parameters (size, shape and depths) of subsurface structures which can be approximated using regular as well as irregular geometries like horizontal cylinder, irregular polygons,

faulted prism, solid sphere, triangles and rectangular lamina (Talwani et al., 1959). Very little attempt has been given to design robust algorithm for the generation of physical and geometrical properties of ore deposits using gravity and gravity gradient together. Dubey and Tiwari (2016) highlighted the influence of density and geometrical parameter using full gravity gradient tensor (FTG) over different synthetic models (Praveen et al., 2022; Sunaina and Dubey, 2022). Usually, vertical gravity field (g_z) provides information of density, location and shape of underneath bodies (Essa, 2007; Hinze et al., 2013; Dubey et. al., 2014; Biswas et al., 2017; Roy et al., 2021a, b), whereas the FTG offers additional knowledge about the location, density and the edges of the body (Saad, 2006; Barraud, 2013; Oliveira and Barbosa, 2013; Oruc et al., 2013; Zhang et al., 2015; Zhou et al., 2015). Therefore, the combination of gravity and gravity gradient inversion (Capriotti and Li, 2014; Qin et al., 2016) improves the estimation of unknown parameters (depth, shape and size, density) using suitable optimization techniques.

However, advanced subsurface imaging in complex geological areas requires effective and robust non-linear optimization due to stochastic nature of source parameters and complex pattern of full gravity gradient tensor. During the last decades, optimization which is featured in almost all aspects of geophysical datasets is becoming a truly indispensable method in many inversion problems. Inversion with forward models using optimization is

ambiguous in nature and incorporating constraints may remove such limitations. For real data inversion, most of the authors used idealized geometric body such as vertical cylinder, horizontal cylinder or spherical geometry with various optimization algorithm. Singh and Biswas (2016) used Particle Swarm Optimization (PSO) method for idealized geometry, Mehanee (2014) used simultaneous regularized inversion, Biswas (2015) used very fast simulated annealing approach while linear and non-linear least square method (Essa, 2011) was used for data errors and maximum likelihood estimator. Similarly, Amjadi and Naji (2013) and Singh and Biswas (2016) used genetic algorithm to improve the efficiency or reduce the expenses. Nevertheless, numerous results are available for the joint inversion of gravity and gravity gradient data sets for shallow surface imaging using irregular polygon but use of PSO is still rare in the field of gravity inversion.

Particle Swarm Optimization (PSO) is increasingly becoming a significant tool to solve stochastic and constrained optimization problem used during the recent years for different aspects of geophysics inversion scheme. It can be effectively applied to find the extremum of nonlinear optimization problems with many independent

parameters like density, susceptibility, depth, location, sizes, and boundaries of any type of source geometries. Recently, PSO is used for source estimation of magnetic anomalies due to simple geometrical structures (Essa and Elhussein, 2018), self-potential idealized bodies for electrochemical, electro-kinetic and thermoelectric sources (Santos, 2010), global PSO for residual gravity anomalies (Singh and Biswas, 2016). However, the use of PSO is less in joint modelling of gravity and full gravity gradient tensor which is still in emerging phase. Joint inversion of gravity and FTG is an attractive work around, and in particular, density model from joint inversion provides a promising solution to shallow subsurface explorations.

Therefore, near surface density models are important for deep subsurface imaging of ore deposits using advanced tool of gravity gradiometry exploration method and other geodynamic applications (Dubey and Tiwari, 2022; Prasad et al., 2021). Such high-resolution imaging also helps in overcoming the problem of potential slope failure during the mining. Though the seismic imaging and electromagnetic method etc are established so far, they may not be directly used due to their high costs and their limitations in complex geological environments of steep geological settings.

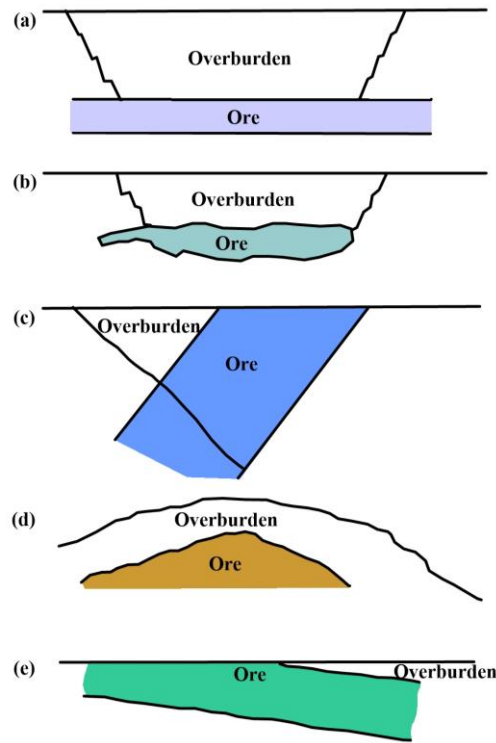


Figure 1. Subsurface ore deposits in different geological scenarios: (a) platinum reefs or coal deposits in form of bed or flat terrain, (b) iron ore or sulphide deposits in form of massive deposits, (c) anthracite deposits in dipping bed, (d) copper sulphide deposits in form of massive deposits and high relief, and (e) iron ore or coal deposits in form of thick bedded with little overburden or flat terrain.

To address such issues, we have developed MATLAB based inversion algorithm using PSO method for joint inversion of gravity and gravity gradient data sets. The aim of the present study is to provide information near/close to the actual values of the ore size, shape and depth for the next stage of exploration. Its specific contributions are:

(i) The developed algorithm with the configuration currently adopted by the exploration community performs well in the context of meaningful scientific analyses applied to synthetic models (regular and irregular shapes)

(ii) Evaluation of joint inversion algorithm on real examples from three different ore deposits such as sulphur, gold and REE that focuses on quantifying differences, which enable intuitive comparisons.

Therefore, an attempt is made to exploit these contributions using joint inversion-based density model with PSO algorithm, in order to image the depth, size, shape and density of ore deposits. The utility of the method is demonstrated using synthetic models followed by three field examples.

METHODOLOGY

A theoretical and synthetic 2.5D gravity and gravity gradient anomaly related to various ore deposits such as triangular, rectangular, polygonal, and spherical body are considered/ taken up to demonstrate the validity and the applicability of the new proposed approach. In the following section, we describe a general technique for the approximation of the parameters of the ore deposits in complex geological settings which is achieved by the successive approximation of multiple nodes to generate different types of simple geometries as well as irregular geometries.

2.5D Gravity and gravity gradient anomaly due to simple geometrical models

The general expressions for 2.5D forward modelling are a requisite for many geophysical applications and useful to compute the gravity and gravity gradient anomalies in local or regional scale geological models. In gravity field analysis, many efforts have been made to improve the efficiency of the forward modelling approach. In cartesian coordinate systems, joint modelling of 2.5D cross section of any y-directional infinitely extent buried mass source in x-z plane can be approximated by a polygon (Wan et al., 2019) as shown in Figure 2. The general gravity (g_z) and gravity gradient (T_{zz} and T_{zx}) effect at an observation point P (x_i, z_i) caused by a polygon having a density contrast ($\Delta\rho$) is given by surface integral and can be written according to Talwani et al. (1959) as:

$$g_z(x_i, z_i) = 2G\Delta\rho \iint_S \frac{(z-z_i)}{(x-x_i)^2 + (z-z_i)^2} dx dz \quad \dots\dots(1)$$

$$T_{zz}(x_i, z_i) = 2G\Delta\rho \iint_S \frac{(x-x_i)^2 - (z-z_i)^2}{((x-x_i)^2 + (z-z_i)^2)^2} dx dz \quad \dots\dots(2)$$

$$T_{zx}(x_i, z_i) = 2G\Delta\rho \iint_S \frac{-2(x-x_i)(z-z_i)}{((x-x_i)^2 + (z-z_i)^2)^2} dx dz \quad \dots\dots(3)$$

Further, this surface integral can be approximated into line integral form using Stoke's law (Zhou, 2009). The final line integral form for gravity field anomaly and gravity gradient at any point P(x_i, z_i) is expressed as:

$$g_z(x_i, z_i) = -2G\Delta\rho \oint \arctan\left(\frac{x-x_i}{z-z_i}\right) dz \quad \dots\dots(4)$$

$$T_{zz}(x_i, z_i) = 2G\Delta\rho \oint \left(\frac{x-x_i}{(x-x_i)^2 + (z-z_i)^2}\right) dz \quad \dots\dots(5)$$

$$T_{zx}(x_i, z_i) = 2G\Delta\rho \oint \left(\frac{z-z_i}{(x-x_i)^2 + (z-z_i)^2}\right) dz \quad \dots\dots(6)$$

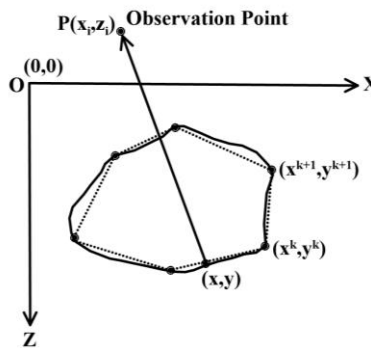


Figure 2. Polygon approximation of 2.5D geometry (solid line represents real model and dotted line shows approximated model to be used for computation).

However, these line integrals can be performed more conveniently using Gaussian quadrature discretization. Thus, forward modelling of gravity and gravity gradients for any generalized body with suitable polygonal approximation can be estimated with input parameter of vertex positions and density of the polygon body and vice versa by suitable optimization techniques through inversion process.

2.5D Joint inversion of gravity and gravity gradient anomaly

The traditional joint inversion objective functional contains the data misfit terms of gravity and gravity gradient data. In the proposed method of joint inversion of gravity and gravity gradient, the objective functional contains three components of the gravity gradiometry data misfit term and the gravity data misfit term along with three constraints (compactness, location and density) the inversion procedure to seek reasonable solutions. Single tensor components, their combinations and/or the standard vertical gravity field component, g_z , can be used in data inversion for geophysical exploration. Different authors explored the performance of single and multiple components in inversion. Biegert et al. (2001) inverted both the gravity field g_z and the FTG components with the aim of retrieving information on minerals deposit for defining shallow structures, whereas g_z allows a better estimation of the source thickness.

Several inversion methods of each data sets are affected by the problem of non-uniqueness due to spatial and temporal limitations of observations. Thus, one often incorporates some constraints into the inversion procedure to seek reasonable solutions. Inversion approaches often use convenient optimization method to estimate the responsible model parameters that provide minimum error between the observed and calculated data sets. But, in general, potential fields like gravity and magnetic contain ambiguity due to the non-uniqueness property since several different configurations of input parameters can generate same output feature. One of the ways to resolve such uncertainties is by incorporating constraints into our optimization problem to force the inverse model into appropriate estimation of input parameters (Dubey et al., 2017 a,b). Such constraints reduce the non-uniqueness issue and can avoid inconsistency between models from individual data inversion. In real scenario of geological model, some prior information of geology, density information from drilling can be used as constraints for given model in joint inversion (Zhdanov, 2009). It reduces the null space for one type of data by the other, and to mutually suppress the influence of different noise sources on individual data sets. Therefore, in global

optimization scheme, model tries to minimize the given misfit functional with suitable choice of parameters as per the given constraints by iterative procedure.

In this present study, we attempt to invert shape and density of any type of subsurface geometry using polygonal approximation. So, the model parameters are location of vertex positions and density (Dubey and Tiwari 2018). The polygonal shape of any regular as well as irregular geometry can be obtained by joining the nearest vertices using straight lines. We have used two different types of polygonal model for generalization of any mineral deposits: (i) 4-point polygon and (ii) 6-point polygon. Therefore, the location of each vertex point consists of two parameters (x and z) for any 2.5D geometric bodies and hence, the total number of parameters is 9 and 13 including density for these three models respectively. In a similar way, any complex ore deposits below the surface can be well approximated by increasing the vertex points to get the better shape estimation although computationally expensive.

Therefore, the gravity and gravity gradient anomaly of discretised of geometry in terms of model parameter values and observation points can be expressed as:

$$g_z^{Model}(x_k, z_k) = f_z(m, x_k, z_k) \dots\dots(7)$$

$$T_{zz}^{Model}(x_k, z_k) = f_{zz}(m, x_k, z_k) \dots\dots(8)$$

$$T_{zx}^{Model}(x_k, z_k) = f_{zx}(m, x_k, z_k) \dots\dots\dots(9)$$

Where f_z , f_{zz} and f_{zx} are the forward discrete functional for evaluating gravity and gradient anomaly at point (x_k, z_k) due to the model parameter values (m). Here, m consists of the vertex position and density of polygon approximation. For any observed gravity and gradient anomaly, the misfit function can be written as:

$$\phi_z(m) = \left(\frac{\sqrt{\frac{1}{N} \sum_{k=1}^N (g_z^{Model}(x_k, z_k) - g_z^{Obs}(x_k, z_k))^2}}{g_z^{Obs(max)} - g_z^{Obs(min)}} \right) \dots\dots(10)$$

$$\phi_{zz}(m) = \left(\frac{\sqrt{\frac{1}{N} \sum_{k=1}^N (T_{zz}^{Model}(x_k, z_k) - T_{zz}^{Obs}(x_k, z_k))^2}}{T_{zz}^{Obs(max)} - T_{zz}^{Obs(min)}} \right) \dots\dots(11)$$

$$\phi_{zx}(m) = \left(\frac{\sqrt{\frac{1}{N} \sum_{k=1}^N (T_{zx}^{Model}(x_k, z_k) - T_{zx}^{Obs}(x_k, z_k))^2}}{T_{zx}^{Obs(max)} - T_{zx}^{Obs(min)}} \right) \dots\dots(12)$$

Thus, total misfit function $\phi(m)$ for joint inversion of gravity and gravity gradient data for a particular set of model parameter m can be written after summing above equations (10), (11) and (12):

$$\phi(m) = \phi_z(m) + \phi_{zz}(m) + \phi_{zx}(m) \dots\dots(13)$$

In the above equations, N is the number of observation points, g_z^{Obs} , T_{zz}^{Obs} , and T_{zx}^{Obs} are the observed gravity and gravity gradient anomaly, whereas g_z^{Model} , T_{zz}^{Model} , and T_{zx}^{Model} are the inverted model anomaly of gravity and gravity gradients.

Further, total misfit function $\phi(m)$ is minimised to estimate the best suitable model parameter (m) using three different types of constraints to reduce the non-uniqueness problem of gravity anomaly as described below:

(i) Compactness constraint (q) i.e. the ratio of the area of an object to the area of a circle with the same perimeter is given by:

$$q = \frac{4\pi \cdot \text{area}}{\text{perimeter}^2} \quad \dots\dots(14)$$

where, q has the most compact shape having minimum value of 1.

(ii) Location constraint i.e. centres position of the body in vertical and horizontal direction.

(iii) Density constraints, a minimum and maximum density barrier.

A new cost function is derived with the constraints using penalty and barrier method into data misfit functional and finally, it is minimized with Particle Swarm Optimization (PSO) technique to achieve optimum model parameter (m) for reconstructing the most desirable shape and density of the given ore deposit.

Particle Swarm Optimization (PSO) algorithmic description

The Particle Swarm Optimization (PSO) is a simple but powerful and intelligent meta-heuristic global optimization technique. It is inspired by the social behaviour of animal such as school of fishes and flock of birds. It is successfully

applied to extensive application in diverse fields of science and engineering like machine learning, image processing, data mining and many other fields. PSO was first introduced by Eberhart and Kennedy (1995) for modelling of social behaviour of animal. PSO contains a population of candidate solutions known as swarm. Each candidate solution in the population is designated as particle and every particle contains the parameter information of the model. Hence, each particle is the candidate solution of the optimization problem. A search space accommodates every possible solution of the optimization problem, and each particle has a position in the search space as shown in Figure 3. Finally, it converges to the best possible solution among all particles in the swarm.

Position of i^{th} particle in search space for any discrete time step can be located by a position $\vec{x}_i(t)$, and a velocity $\vec{v}_i(t)$ of same dimension and represents the movement of particles in each time steps. Further, each particle has its best position ($\vec{p}_i^{best}$) where misfit function for this particular particle is minimum throughout the discrete time steps. In addition, a global best position also exists for the whole swarm and that can be symbolized as ($\vec{g}^{best}$). In moving direction towards the best solution, the position and the velocity of each particle regularly gets updated after each stage of movement/iteration towards the combination of direction of ($\vec{p}_i^{best}$), ($\vec{g}^{best}$) and previous velocity of the particle. The whole procedure of swarm movement is graphically represented in Figure 3. The updated velocity and position for any individual particle at time step (t) can be mathematically written as:

$$\vec{v}_i(t+1) = w\vec{v}_i(t) + c_1r_1(\vec{p}_i^{best}(t) - \vec{x}_i(t)) + c_2r_2(\vec{g}^{best}(t) - \vec{x}_i(t)) \quad \dots\dots(15)$$

$$\vec{x}_i(t+1) = \vec{x}_i(t) + \vec{v}_i(t+1) \quad \dots\dots(16)$$

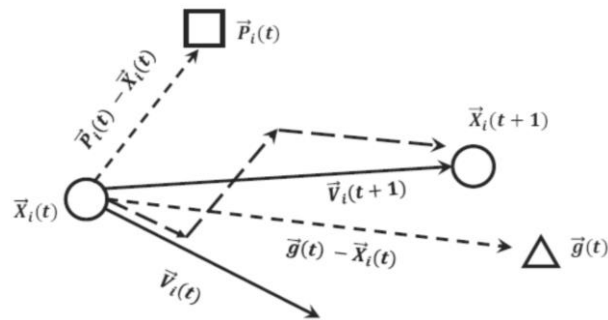


Figure 3. Vector representation of position and velocity of a particles from swarm for Particle Swarm Optimization (PSO) computation.

Here, $w\vec{v}_i(t)$ is known as inertia term, c_1 and c_2 are the acceleration coefficients, r_1 and r_2 are two random numbers in range [0 1], $c_1r_1(\vec{p}_i^{best}(t) - \vec{x}_i(t))$ is known as cognitive component and $c_2r_2(\vec{g}^{best}(t) - \vec{x}_i(t))$ is known as social component. The new velocity vector can translate the position vector into a new location of the search space for the sake of optimization. The values of co-efficient constraint vary between $0 < c_1 + c_2 < 4$ and $(\frac{c_1+c_2}{2} - 1) < w < 1$ for stable optimization (Perez and Behdinan, 2007). In this present study, following values are considered for each iteration during the inversion process:

$$c_1 = 1 \text{ \& } c_2 = 2 \dots\dots(17)$$

$$w(t+1) = 0.1 < w(t) * 0.9 < 1 \dots(18)$$

A flow chart for the development of PSO algorithm used in the present study is described in Figure 4.

PROBLEM VISUALIZATION AND SOLUTION VALIDATION: SYNTHETIC EXAMPLES

To verify the validity and feasibility of our proposed PSO based joint inversion of the gravity and gravity gradient data, we tested it using a complex combined model that is more similar to the real underground structure in the form of different types of ore deposits along the faults or other geological elements. Three different anomalous bodies are separately embedded in the uniform half-space. Anomalous bodies like irregular prism, triangulated mass and spherical mass are considered as isolated ore deposits of different sizes, shapes and density at different burial depths (Figures 5, 7, and 9). The joint inversion for these synthetic models is carried out with 4-points, and 6-points polygonal approximations using PSO algorithm for their original location, shape, size, depth and density estimation.

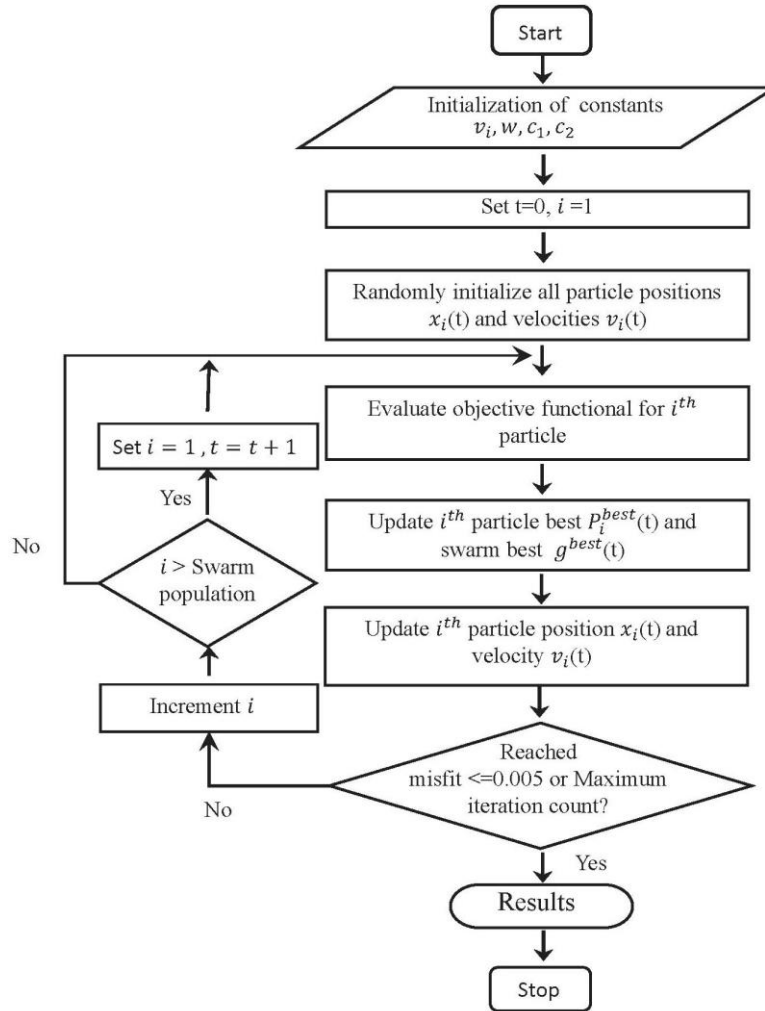


Figure 4. Flow chart of Particle Swarm Optimization (PSO) algorithm.

Each density model adopts the same mesh size (2000×1000) in the joint inversion with same observation points of gravity and gravity gradients. There are total of 200 observation points with a constant sample spacing of 10 m from a horizontal distance of 0 to 2000 m along a profile. To simulate the noisy data, a 10% random Gaussian noise was added to the gravity and gravity gradients forward model data sets that might arise during measurements and processing of raw gravity and gravity gradient data sets or small interferences due to nearby sources. Two different types of inversion approach were adopted viz. (1) shape optimization with fixed density, and (2) both shape and varying density optimization for real and noisy datasets. Therefore, a total of four different cases of inversion is performed with 40 test runs for each set of synthetic model and mean model parameters are evaluated.

The joint inversion results of gravity (g_z) and gravity gradients (T_{zz} , T_{zx}) are discussed in the next section for the above described synthetic models to estimate the detailed architecture of any irregular shape of ore deposits in terms of overlapping areas (shape and size), excess areas, location, depth and density although the actual parameters are in the form of vertex position and density. We have used following outputs to describe the results of joint inversion for better perceptive:

- (i) Percentage of common overlapping area of actual and inverted shape
- (ii) Percentage of excess area from actual shape

(iii) Centre position of the ore body

(iv) Depth of the ore body

(v) Density of the ore body

These above five inverted outputs from the joint inversion for all the above models are also discussed in terms of histogram plot between model parameters and most probable frequency distribution correspondingly. These parameters are characterized with 10 bins and range of bins is 10 equally distributed points from maximum and minimum values of each parameters. Here, we discuss the joint inversion results in detail for each model one after the other

Irregular prism or laminar ore deposits

Such synthetic model is usually approximated to model the several anomalous zones like any type of fault, sheets, dykes or sills etc., where the possibilities of ore deposits are enhanced. The initial and estimated model parameters are given in Tables 1a and 1b. Results of joint inversion of gravity and gravity gradient using 4-point, and 6-point polygon for four different cases are shown in Figure 5a, b that illustrates the precise estimation of the size, shape, depth and density of the irregular prism having real as well as noisy data sets. However, the histogram plots of the frequencies distribution of estimated parameters are shown in Figures 6 and 7 for both cases of fixed as well as varying density model.

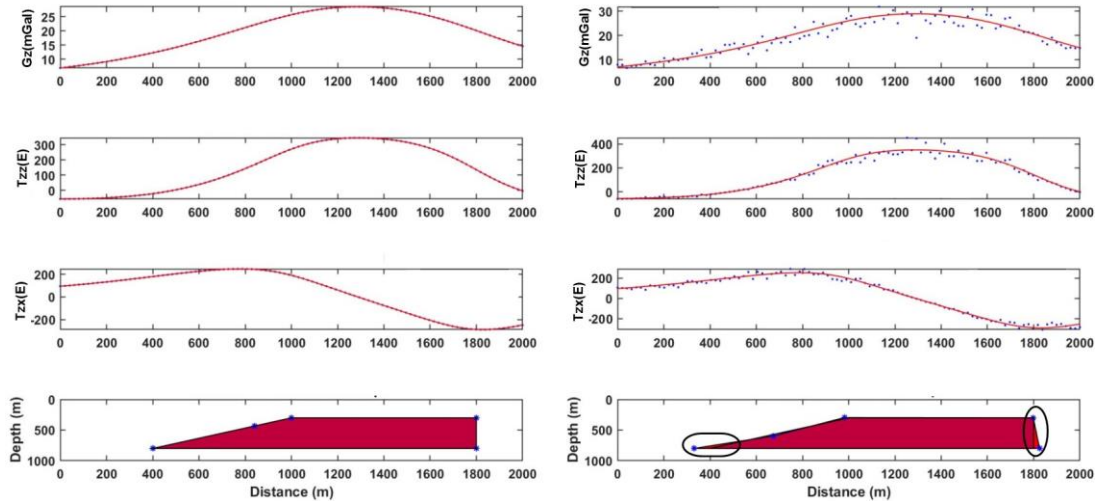


Figure 5. Joint inversion results of 2.5D irregular prism in terms of fault deposits where solid red lines represent computed G_z , T_{zz} and T_{zx} anomaly and blue dotted line show inverted G_z , T_{zz} and T_{zx} anomaly respectively. **(a)** Model without noise, **(b)** Model with noise.

In case of fixed density model, the estimated model parameters like overlapping areas are very close to 100% (98 - 100), probability distribution of excess areas are near to 0 to 1%, location of the model is 1236.30 m almost nearer to the initial value of 1236.36 m, and depth to the centre of model is 572 m close to the initial value of 572.73 m, obtained using real data sets of gravity and gravity gradients without noise as shown in Figure 6a. However, similar set of these parameters having a small variation are also achieved using noisy data sets with fixed density model as shown in Figure 6b.

On the other side of computation with varying density model, the estimated model parameters are highlighted in

the Figure 7a, b with noise free and noisy data sets respectively. From the histograms plot shown in Figure 7, it is obvious to note that the estimated model parameters are very close to the initial values but not as compared to the results from the model with fixed density computation discussed above in Figure 6. Such deviations in model parameters occur due to the noise presence in the real data sets and can be easily seen in Figure 5b. The final density model in all the four different cases are obtained with the data misfit from 4-points, and 6-points polygon as shown in Table 1 with all other estimated model parameters whose confidence limit is almost 95% or more.

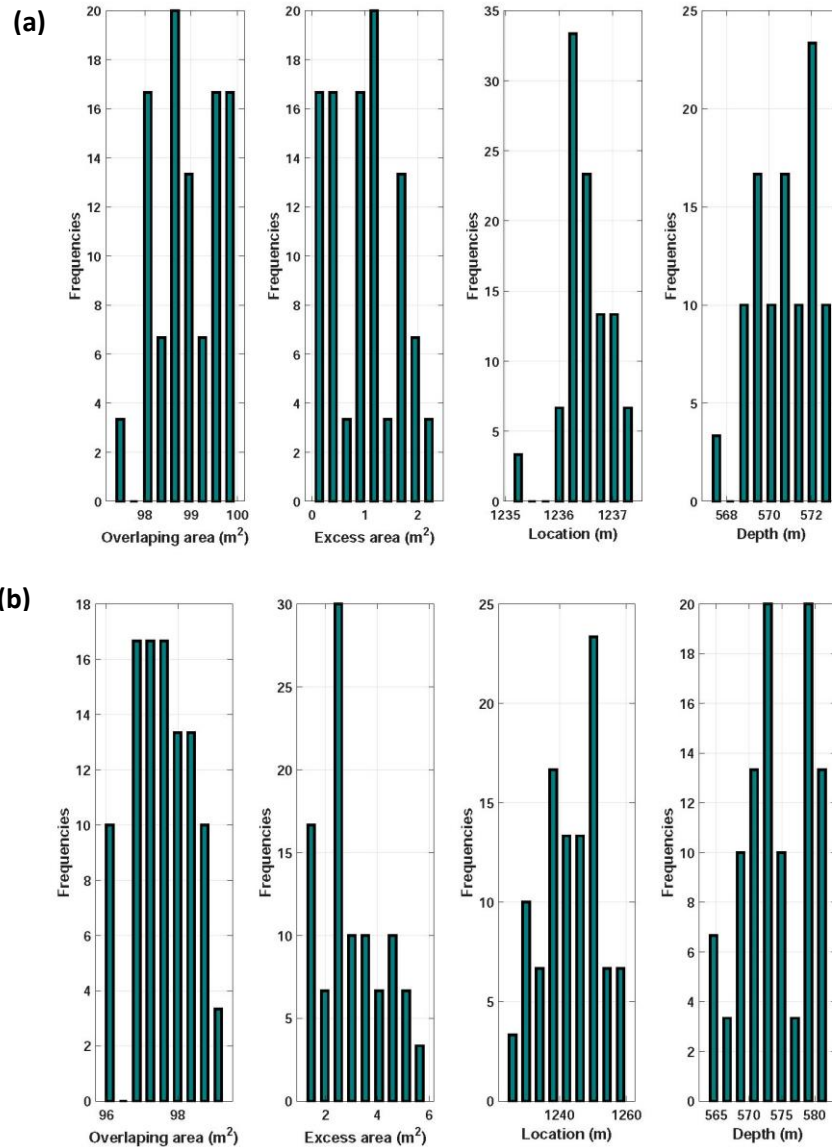


Figure 6. Frequency distribution of inverted model parameters for 2.5D irregular prism with fixed density. (a) Inversion results for model without noise, (b) Inversion results for model with noise.

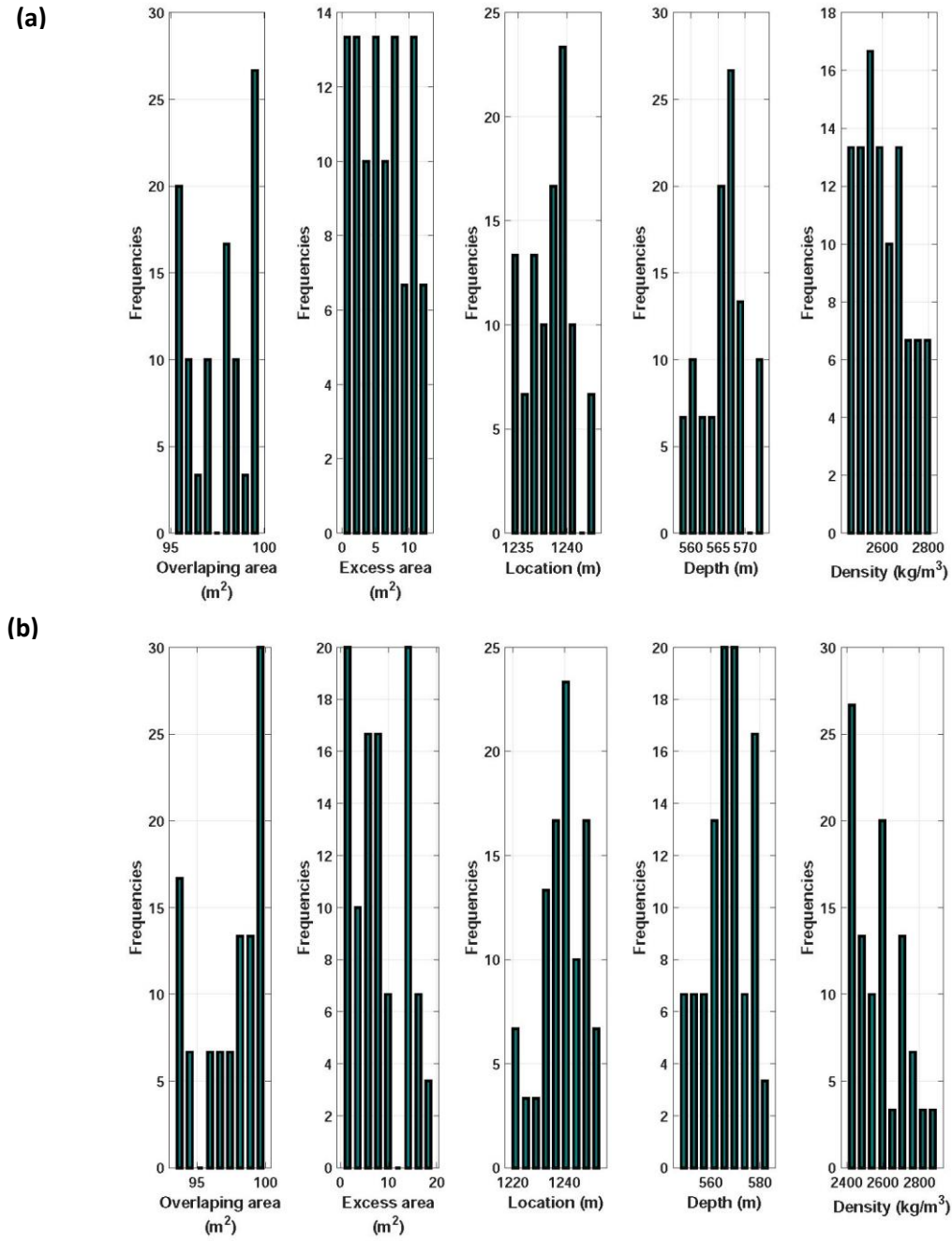


Figure 7. Frequency distribution of inverted model parameters for 2.5D irregular prism with varying density. **(a)** Inversion results for model without noise, **(b)** Inversion results for model with noise.

Table 1a. Parameters for irregular prism model with 4-point polygonal shape and density inversion

Model Parameters	Actual Values	Search Range	fixed density (without noise)	varying density (without noise)	fixed density (with 10% noise)	varying density (with 10% noise)
Common Area (%)	100	-	98.87 ± 0.79	97.195 ± 1.70	96.18 ± 1.78	95.130 ± 3.10
Excess Area (%)	0	-	1.02 ± 0.71	3.288 ± 3.41	3.81 ± 1.90	4.115 ± 3.78
x position (m)	1236.36	0 to 2000	1236.32 ± 0.07	1237.00 ± 1.77	1240.59 ± 7.53	1236.44 ± 11.18
z position (m)	572.73	0 to 1000	570.51 ± 1.57	568.25 ± 4.77	568.35 ± 6.65	567.43 ± 10.47
Density (kg/m^3)	2700	1000 to 3000	2700	2684.34 ± 93.86	2700	2719.93 ± 148.77
Misfit %	-	-	0.013 ± 0.01	0.043 ± 0.03	6.52 ± 0.12	7.70 ± 0.04

Table 1b. Parameters for irregular prism model with 6-point polygonal shape and density inversion

Model Parameters	Actual Values	Search Range	fixed density (without noise)	varying density (without noise)	fixed density (with 10% noise)	varying density (with 10% noise)
Common Area (%)	100	-	96.89 ± 1.63	97.29 ± 1.63	95.09 ± 1.64	96.25 ± 2.84
Excess Area (%)	0	-	2.76 ± 1.43	8.45 ± 3.70	4.69 ± 1.57	10.34 ± 4.99
x position (m)	1236.36	0 to 2000	1236.07 ± 2.97	1237.96 ± 2.95	1237.36 ± 10.73	1239.49 ± 9.48
z position (m)	572.73	0 to 1000	566.80 ± 3.60	562.45 ± 4.58	565.76 ± 6.09	561.61 ± 10.55
Density (kg/m^3)	2700	1000 to 3000	2700	2544.75 ± 102.70	2700	2530.02 ± 137.28
Misfit %	-	-	0.065 ± 0.10	0.094 ± 0.05	8.45 ± 0.13	7.06 ± 0.05

Triangulated ore deposits

Triangular geometries are another type of simple geometry that is more accurate and precise approximation of any irregular subsurface geometries or interfaces. Joint gravity and gravity gradient forward models have been estimated and also simulated the results with 10% Gaussian noise for polygon shape and density inversion as shown in Figure 8 a, b. All four types of inversion as described in earlier section along with frequency statistics are evaluated but we have shown only with and without noise varying density and shape optimization parameter statistics. The standard

deviation of all the models with 4-points and 6-points polygon approximation and mean values of estimated parameters are shown in Figure 9. Here, initial and estimated model parameters for joint inversion of gravity gradient using 4-point and 6-points polygon approximation is given in Tables 2a and 2b respectively. Joint inversion results show almost similar results obtained for 5-points and 6-point polygon as shown in Table 2. It is interesting to note that the fixed density model parameters are more confined with less standard deviation unlike previous model for 2.5D irregular prism model.

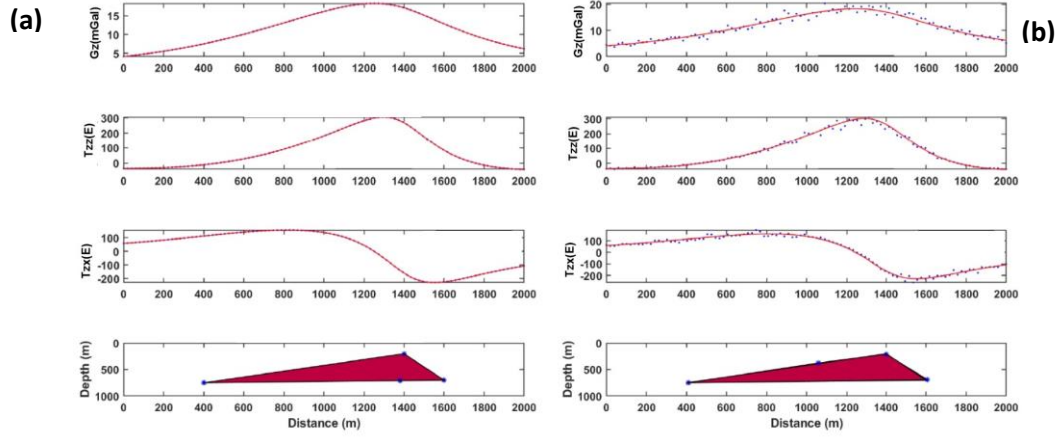


Figure 8. Joint inversion results of 2.5D triangle in terms of any irregular ore deposits where solid red lines represent computed Gz, Tzz and Tzx anomaly and blue dotted line show inverted Gz, Tzz and Tzx anomaly respectively. **(a)** Model without noise, **(b)** Model with noise.

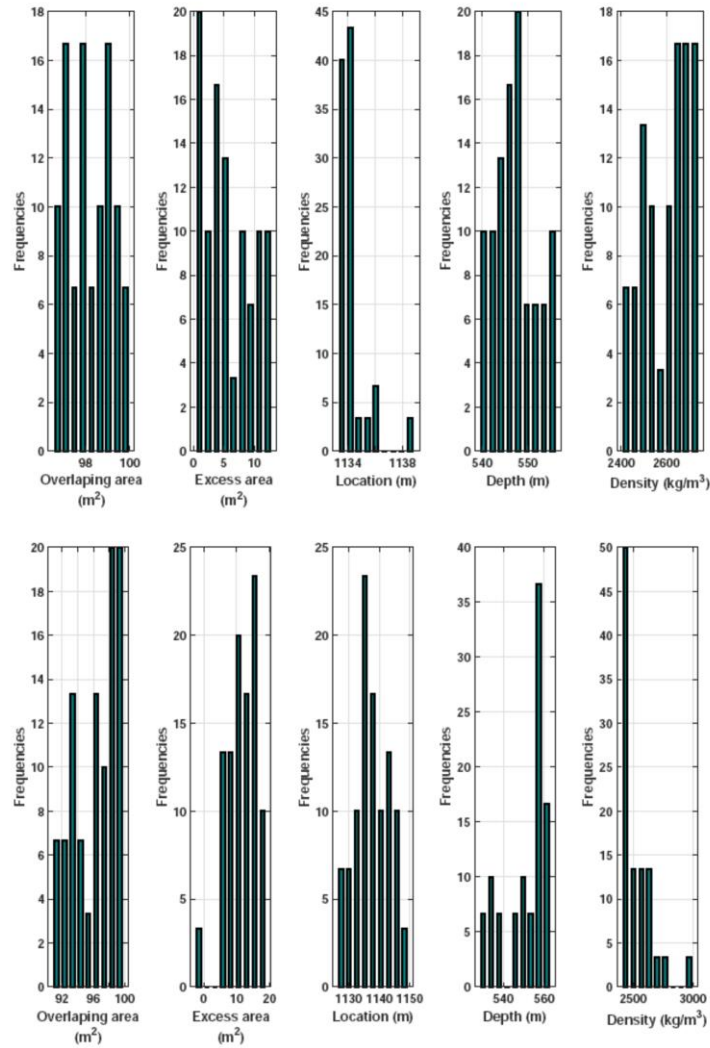


Figure 9. Frequency distribution of inverted model parameters for 2.5D triangle with varying density. **(a)** Inversion results for model without noise, **(b)** Inversion results for model with noise.

Table 2a. Parameters for the model of triangulated geometry with 4-point polygonal shape and density inversion

Model Parameters	Actual Values	Search Range	fixed density (without noise)	varying density (without noise)	fixed density (with 10% noise)	varying density (with 10% noise)
Common Area (%)	100	-	98.59 ± 1.10	97.31 ± 1.93	94.39 ± 1.98	93.77 ± 5.09
Excess Area (%)	0	-	1.31 ± 1.03	5.68 ± 4.03	5.49 ± 2.09	11.24 ± 4.89
x position (m)	1133.33	0 to 2000	1133.43 ± 0.53	1134.50 ± 1.38	1135.50 ± 5.20	1136.93 ± 5.96
z position (m)	550.00	0 to 1000	548.72 ± 1.93	546.55 ± 4.81	549.53 ± 8.29	548.05 ± 12.87
Density (kg/m^3)	2700	1000 to 3000	2700	2619.99 ± 110.74	2700	2577.50 ± 89.56
Misfit %	-	-	0.025 ± 0.02	0.069 ± 0.05	6.44 ± 0.07	6.91 ± 0.05

Table 2b. Parameters for the model of triangulated geometry with 6-point polygonal shape and density inversion

Model Parameters	Actual Values	Search Range	fixed density (without noise)	varying density (without noise)	fixed density (with 10% noise)	varying density (with 10% noise)
Common Area (%)	100	-	90.59 ± 4.47	86.29 ± 5.22	88.126 ± 4.60	87.35 ± 6.98
Excess Area (%)	0	-	8.99 ± 4.12	18.27 ± 8.02	11.05 ± 3.97	18.65 ± 7.87
x position (m)	1133.33	0 to 2000	1135.55 ± 8.22	1139.99 ± 5.54	1136.86 ± 8.26	1138.77 ± 9.18
z position (m)	550.00	0 to 1000	541.29 ± 9.99	528.02 ± 5.54	539.27 ± 13.54	529.38 ± 17.17
Density (kg/m^3)	2700	1000 to 3000	2700	2550.37 ± 128.05	2700	2518.19 ± 142.32
Misfit %	-	-	0.247 ± 0.72	0.295 ± 0.19	7.45 ± 0.05	6.68 ± 0.09

Spherical deposits or massive ore deposits

This is another example of synthetic model of massive sulphide deposits which can be generally seen in the form of almost spherical shape. Therefore, we have adapted vertex points and density for forward gravity and gravity gradient evaluation. 2.5D spherical geometry can be constructed as an infinite number of equidistant points from a centre position. In the present case, we have used 100 equidistant vertex points from the centre position for inverting the shape, size, depth, location and density of the geometry with

4-point and 6-point polygon approximations. The results of joint inversion and estimated model parameters are shown in Figures 10 and 11. Similar configuration and set up as described in earlier sections were taken for the computation and the results can be discussed in same manner. However, it is interesting to observe that the best approximations with less standard deviation of estimated model parameters from initial values can be monitored for the 4 and 6-point polygon approximations as given in Tables 3a and 3b. These results are consistent for noisy, fixed density and varying density models.

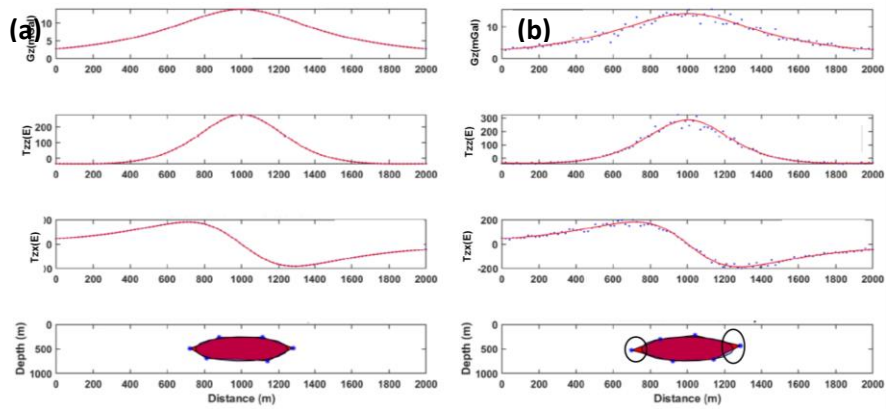


Figure 10. Joint inversion results of 2.5D sphere in terms of massive ore deposits where solid red lines represent computed Gz, Tzz and Tzx anomaly and blue dotted line show inverted Gz, Tzz and Tzx anomaly respectively. **(a)** Model without noise, **(b)** Model with noise.

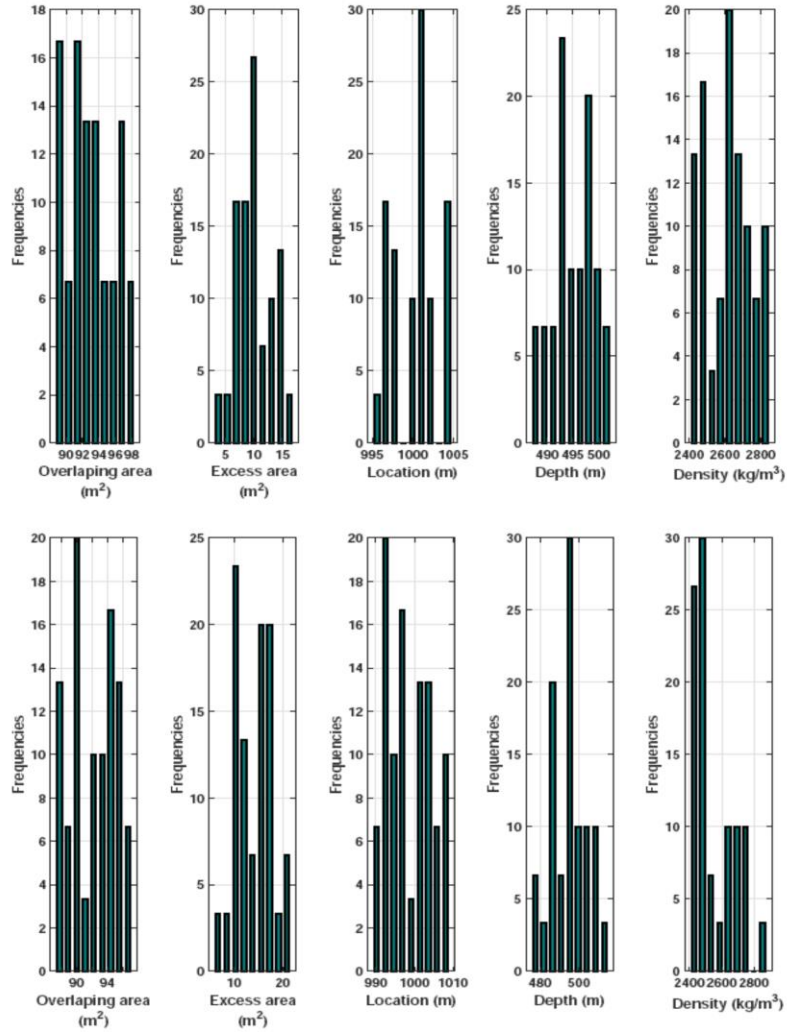


Figure 11. Frequency distribution of inverted model parameters for 2.5D sphere with varying density. **(a)** Inversion results for model without noise, **(b)** Inversion results for model with noise.

Table 3a. Parameters for the spherical model with 4-point polygonal shape and density inversion

Model Parameters	Actual Values	Search Range	fixed density (without noise)	varying density (without noise)	fixed density (with 10% noise)	varying density (with 10% noise)
Common Area (%)	100	-	90.18 ± 0.15	84.92 ± 2.14	90.05 ± 0.52	86.02 ± 2.63
Excess Area (%)	0	-	10.36 ± 1.2	6.51 ± 1.66	10.25 ± 0.70	7.32 ± 2.12
x position (m)	1000	0 to 2000	1000.57 ± 1.33	1000.27 ± 1.74	1000.07 ± 5.51	999.84 ± 4.09
z position (m)	500	0 to 1000	503.64 ± 2.31	497.63 ± 6.11	503.14 ± 6.56	498.86 ± 9.23
Density (kg/m^3)	2700	1000 to 3000	2700	2952.66 ± 99.78	2700	2902.15 ± 135.36
Misfit %	-	-	0.223 ± 0.04	0.227 ± 0.09	6.471 ± 0.05	6.21 ± 0.08

Table 3b. Parameters for the spherical model with 6-point polygonal shape and density inversion

Model Parameters	Actual Values	Search Range	fixed density (without noise)	varying density (without noise)	fixed density (with 10% noise)	varying density (with 10% noise)
Common Area (%)	100	-	92.11 ± 0.92	93.01 ± 2.79	91.02 ± 1.25	92.13 ± 3.04
Excess Area (%)	0	-	7.44 ± 0.78	10.05 ± 3.24	8.98 ± 1.46	14.03 ± 3.83
x position (m)	1000	0 to 2000	999.79 ± 1.57	1000.14 ± 2.76	1000.51 ± 5.00	998.52 ± 5.70
z position (m)	500	0 to 1000	495.31 ± 4.22	495.07 ± 3.90	498.47 ± 6.50	494.14 ± 9.67
Density (kg/m^3)	2700	1000 to 3000	2700	2614.21 ± 129.82	2700	2544.71 ± 134.28
Misfit %	-	-	0.137 ± 0.09	0.162 ± 0.10	5.72 ± 0.04	5.63 ± 0.06

Structure identification and surface refinement

Notably, the gravity inversion alone fails to recover the true shape geometry and position of the anomalous bodies, and the vertical resolution of the inversion results is very poor, and it is difficult to accurately identify the varying density anomalous bodies. Moreover, the inversion result of the gravity alone still has problems, such as the upset of the centre of the anomalous bodies, the blurring of the boundary between the anomalous bodies and the surrounding rock, and the inability to recover the exact physical parameter value. However, the gravity gradient data sets reflect the spatial position and geometry of the true anomalous bodies. Compared with the results of the gravity inversion, the distribution of the recovered underground anomaly can be improved. The joint inversion with gravity and gravity gradient (Figures 5, 8, and 10) recovers the gross structure of the synthetic model and their densities. The structural

features of the anomalous bodies in all the three different cases of synthetic models can be clearly distinguished. The inverted model produces the clear boundaries of anomalous bodies in addition to its geometrical and physical parameters.

APPLICATION TO REAL FIELD DATA

We demonstrate the practical utility of the proposed method to estimate the size, shape, depth and density from three field examples of gravity and gravity gradient data for which source parameters are known from the previous results for the validation of the present results as discussed below.

Interpretation of the salt dome

Residual gravity anomaly data from offshore Louisiana salt dome, USA has been taken from Nettleton (1976) and Roy

et al. (2000) which is a famous geological site for oil and gas potentials. Data is digitized for gravity anomaly and further, horizontal gravity gradient (T_{zx}) data is generated from 82 data points along a profile length of 17 km as shown in Figure 12a. Previous study by Mehanee (2014) used spherical, vertical and horizontal model for single data inversion and found the depth to be 2874 meter. Similar study on same data sets was performed by Singh and Biswas (2016) with GPSO optimization method used for estimating the different model parameters like density, depth and radius calculation.

In this study, we have used presently developed PSO based joint inversion of gravity and gravity gradients to estimate the accurate size, shape, depth, location as well as density of the anomalous zone. All results from the inverted model are

shown in Table 4 and compared with the earlier results. The results are consistent for all three types of polygonal approximations.

Interpretation of the sulphide ore deposits

Another real residual gravity profile of sulphide ore body of profile length 300 m (Roy et al. 2000) from Noranda Mining District, Quebec, Canada has been considered for joint inversion of gravity and gravity gradient. Horizontal gravity gradient (T_{zx}) anomaly is generated by taking horizontal derivative of digitized gravity profile with 78 data points as shown in Figure 12b. Mehanee (2014) estimated ~47 m depth of the source along the profile using horizontal cylinder approximation for the sulphide ore deposit.

Table 4. Obtained parameter values of the offshore Louisiana Salt Dome, USA from joint inversion of gravity and gravity gradients data.

Model Parameters	Search Range	Mehanee (2014)	Singh (2015)	Present work		
				5-point polygon	6-point polygon	7-point polygon
x position (m)	-2000 to 2000	-	451.02 ± 24.63	371.26 ± 6.46	370.01 ± 9.89	287.147 ± 92.73
z position (m)	0 to 5000	2899	2874.80 ± 53.90	2929.11 ± 12.19	2924.01 ± 15.06	2773.03 ± 208.67
Density (kg/m^3)	-4000 to 0	-	-2510 ± 1210	-2145.13 ± 195.74	-2082.14 ± 165.54	-2053.27 ± 103.34
Area (m^2)	-	-	598382 ± 53019	586842 ± 46873	602392 ± 37052	588393 ± 34638
Misfit %	-	-	6.7±15	6.73±0.02	6.75±0.03	7.97±1.98

Table 5. Obtained parameter values of the Noranda Mining District, Quebec, Canada from joint inversion of gravity and gravity gradients data.

Model Parameters	Search Range	Mehanee (2014)	Singh (2015)	Present work		
				5-point polygon	6-point polygon	7-point polygon
x position (m)	-150 to 150	-	2.37 ± 0.41	0.43 ± 0.32	0.61 ± 0.41	0.59 ± 0.38
z position (m)	0 to 125	47	46.69 ± 0.47	53.74 ± 0.61	55.06 ± 0.97	54.64 ± 1.16
Density (kg/m^3)	0 to 4000	-	2430.00 ± 950	2424.64 ± 130.52	2442.03 ± 135.82	2442.22 ± 143.75
Area (m^2)	-	-	2748.81 ± 135.50	2513.07 ± 138.41	2537.60 ± 150.63	2527.93 ± 153.97
Misfit %	-	-	3.04 ± 5.2	9.05 ± 0.05	8.84 ± 0.18	8.86 ± 0.19

However, we have used the present PSO algorithm based joint inversion of gravity and gravity gradient data to estimate the overall information about the ore deposit along the same profile. The results of joint inversion with 6-point polygon approximation is shown in Figure 12b and a comparison table with previous results for location, area and density is shown in Table 5. The estimated model parameters of sulfide ore deposits show significant improvement produced by presently developed joint inversion algorithm and corroborate well with previous published results.

Interpretation of the REE ore deposits

Recently, REE (Rare Earth Elements) deposits are in global demands for the mineral resources, like pyrrhotite ore

deposits. Therefore, a residual gravity profile of length around 26 m is taken from Karrbo, Vastmanland, Sweden (Shaw and Agarwal, 1990) to demonstrate the present joint inversion algorithm. Horizontal gravity gradient anomaly (T_{zx}) is generated from the digitized data for joint inversion as shown in Figure 12c. The estimated parameters of such REE deposits from joint inversion with PSO based algorithm shows the location of the deposits with 6-points polygon is to be around 4.82 meter from the surface. The inverted depth of deposit is consistent with the result of 4.82 m obtained by Tlas et al. (2005), while Shaw and Agarwal (1990) estimated ~5.8 meter of source depth using horizontal cylindrical source. The estimated model parameters and their comparison with previous studies are given in Table 6.

Table 6. Obtained parameter values of the Karrbo, Vastmanland, Sweden from joint inversion of gravity and gravity gradients data.

Model Parameters	Search Range	Tlas et al. (2005)	Singh (2015)	Present work		
				5-point polygon	6-point polygon	7-point polygon
x position (m)	-15 to 15	0.18	0.19 ± 0.03	0.22 ± 0.05	0.21 ± 0.04	0.20 ± 0.07
z position (m)	0 to 20	4.82	4.69 ± 0.03	4.82 ± 0.08	4.81 ± 0.08	4.85 ± 0.09
Density (kg/m^3)	0 to 10000	-	9500.00 ± 3260	8494.27 ± 1145.22	7806.49 ± 1041.03	7497.93 ± 1135.87
Area (m^2)	-	-	8.24 ± 2.16	44.29 ± 5.74	48.07 ± 5.83	50.37 ± 6.63
Misfit %	-	-	1.00 ± 2.0	5.46 ± 0.03	5.47 ± 0.03	5.45 ± 0.04

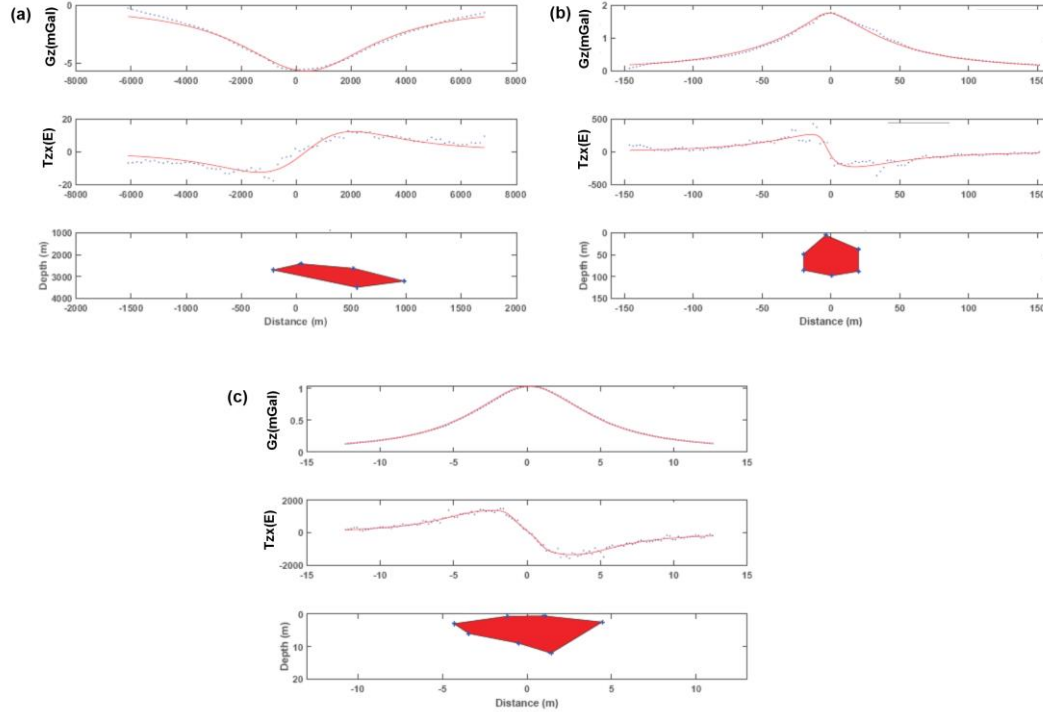


Figure 12. Application of joint inversion results of three real examples from different region of ore deposits. **(a)** Offshore Louisiana Salt Dome, USA; **(b)** Noranda Mining District, Quebec, Canada; **(c)** Karrbo, Vastmanland, Sweden where solid red lines represent observed G_z , and T_{zx} anomaly and blue dotted line show inverted G_z and T_{zx} anomaly respectively.

CONCLUSION

We developed a MATLAB based numerical approach to the joint inversion of gravity and gravity gradient data based on Particle Swarm Optimization method using for polygon approximations as a number of vertex points. Geological evidences envisage complex irregular shape of subsurface bodies and modelling them with polygonal approximation is more realistic than simple regular shaped approximation. Therefore, selecting number of parameters for an optimum model to minimize the misfit functional with optimum polygon vertex is tedious. The use of the developed algorithm for joint inversion using PSO method has been illustrated by synthetic model of different type of ore deposits for the first time. Further, we verified the developed inversion algorithm over three real examples from, (i) Offshore Louisiana Salt Dome, USA, (ii). Noranda Mining District, Quebec, Canada, (3) Karrbo, Vastmanland, Sweden, for its robustness and effectiveness. The result of joint inversion of gravity and gravity gradient is more conducive to geological interpretation. The inferences from the real data examples facilitate the clear and meticulous identification of the subsurface structural information that is always useful in hydrocarbon or mineral explorations. The results of joint inversion in structural classification will lead

to improved subsurface characterization in complicated geological settings for the different type of ore deposits.

Therefore, presently developed MATLAB based algorithm for joint inversion of gravity and gravity gradient with polygonal approximation has the ability to decipher underneath structure, area, location and density with suitable implementation of constraints and is compatible with all available geological information. The developed algorithm can be further extended towards deep subsurface minerals and hydrocarbon exploration.

Data and code availability

In this present study, we have used PSO for inverting underneath structure and density from residual gravity and gravity gradient anomaly. Total 40 tests have been performed for evaluating mean parameter values (i.e. vertex positions and density). Population size or swarm size has been taken 50 having maximum iterations 300 with data misfit stopping criterion 0.005. A processor Intel(R) Xeon-W-145 CPU @3.70 GHZ was used for computations and approximately 10.61 sec is required for 13 parameters model with 300 iterations using MATLAB (R2018a) environment. The code can be found from the GITHUB (<https://github.com/prakashnccess/PSOPA>).

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Compliance with Ethical Standards

The authors declare no conflict of interest and adhere to copyright norms.

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